# A Mechanical Characterization of Conformal Prediction and its Irreducible Excess Risk

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**Abstract**

Every split, full, normalized, Mondrian, quantile, and predictive-system conformal scheme produces the same shape of object: a frozen monotone reshaping of a label-blind, stratum-pooled empirical law. We make that statement mechanical, so it can be checked on any scheme by naming two ingredients, a score map and a stratification. We then show that the excess logarithmic risk of any such scheme over the conditional oracle equals, in the large-calibration limit, a single conditional mutual information between the nonconformity score and the input given the stratum. The quantity depends only on the two frozen ingredients, so no amount of calibration reduces it. It is positive unless the frozen ingredients already render the score conditionally independent of the input, that is, unless the modeling was finished before the conformal step. The method's stated use case, doubt that the residual law is conditionally adequate, is exactly the condition under which its floor is positive. A corollary recovers the regressogram rate handicap of universally consistent conformal predictive systems from the same constraint.

## 1 Setup

Let $(X, Y)$ have joint law $P$ on $\mathcal{X} \times \mathbb{R}$, with conditional law $P(\cdot \mid X)$ and conditional density $p(\cdot \mid x)$. The benchmark is the conditional oracle $P(\cdot \mid X)$, whose expected logarithmic loss is the conditional entropy $H(Y \mid X)$. For a predictive law $F = \{F_x\}$ with density $f_x$ we score by the logarithmic rule and measure a forecaster by its *excess risk*

$$L(F) - H(Y \mid X) \;=\; \mathbb{E}_X\big[\,\mathrm{KL}\big(P(\cdot \mid X)\,\big\|\,F_X\big)\,\big] \;\geq\; 0, \tag{1}$$

the average Kullback–Leibler divergence from the truth to the forecast. Excess risk is zero if and only if the forecaster is the oracle almost surely.

A conformal scheme is fixed by two objects and a fold. A *nonconformity score* $A : \mathcal{X} \times \mathbb{R} \to \mathbb{R}$ assigns a scalar $A(x, y)$ to a candidate value $y$ at $x$. A *stratification* $\kappa : \mathcal{X} \to \mathcal{K}$ assigns each $x$ to a category. Both are computed from a training fold that is disjoint from the calibration data. The calibration multiset $D_n = \{Z_i = (X_i, Y_i)\}_{i=1}^n$ enters only through the pooled scores. Write $\widehat{G}_\kappa$ for the empirical distribution of the calibration scores in stratum $\kappa$,

$$\widehat{G}_\kappa \;=\; \text{empirical law of } \{\, A(Z_i) : \kappa(X_i) = \kappa \,\}.$$

## 2 The invariant

**Definition 1** (Conformal predictor)**.** The conformal predictive distribution function at $x$ is

$$\widehat{F}_x(y) \;=\; \widehat{G}_{\kappa(x)}\big(A(x,y)\big). \tag{2}$$

Its prediction set at level $\alpha$ is the sublevel set $\{\, y : A(x,y) \le \widehat{Q}_{1-\alpha}(\kappa(x)) \,\}$, where $\widehat{Q}_{1-\alpha}(\kappa)$ is the $(1-\alpha)$ quantile of $\widehat{G}_\kappa$.

Definition 1 is the split, Mondrian, normalized, quantile, and predictive-system output written once; full conformal is the case in which $A$ depends symmetrically on the augmented multiset, treated in Section 5. Each named scheme is obtained by naming $(A,\kappa)$: split conformal uses $A(x,y) = |y - \widehat{\mu}(x)|$ with one stratum; locally weighted or normalized conformal uses $A(x,y) = |y - \widehat{\mu}(x)|/\widehat{\sigma}(x)$; conformalized quantile regression uses $A(x,y) = \max\{\widehat{q}_{\mathrm{lo}}(x) - y,\; y - \widehat{q}_{\mathrm{hi}}(x)\}$; Mondrian and group-conditional conformal keep the score simple and make $\kappa$ a fixed partition; the universally consistent conformal predictive system of Vovk (2022) uses a fixed dyadic partition $\kappa = \kappa_n$ with cells shrinking in $n$.

**Lemma 2** (Reshaping form)**.** *Suppose $A(x,\cdot)$ is a strictly increasing bijection of $\mathbb{R}$ for each $x$. Then the law $\widehat{F}_x$ of* (2) *is the pushforward of the pooled law $\widehat{G}_{\kappa(x)}$ through $A(x,\cdot)^{-1}$,*

$$\widehat{F}_x \;=\; A(x,\cdot)^{-1} \circ \widehat{G}_{\kappa(x)}.$$

*Proof.* Immediate from (2): $\widehat{F}_x(y) = \widehat{G}_{\kappa(x)}(A(x,y))$ is the distribution function of $A(x,\cdot)^{-1}(W)$ for $W \sim \widehat{G}_{\kappa(x)}$. □

The content of Lemma 2 is that in every conformal scheme the input $x$ touches the predictive law in exactly two frozen ways: through the deterministic reshaping $A(x,\cdot)$, and through the choice of which stratum's pooled empirical law to use. All non-location content of the forecast, its shape, is a single pooled object $\widehat{G}_{\kappa(x)}$, common to every $x$ in a stratum.

## 3 The excess-risk floor

**Assumption 1** (Frozen design)**.** The pair $(A,\kappa)$ is measurable with respect to the training fold and independent of the calibration labels $\{Y_i\}$.

**Assumption 2** (Regularity)**.** For each $x$, $A(x,\cdot)$ is a strictly increasing, absolutely continuous bijection $\mathbb{R} \to \mathbb{R}$ with derivative $A'(x,y) = \partial_y A(x,y) > 0$. The laws $P(\cdot\,|\,x)$ and the stratum-pooled score laws admit densities, and the divergences below are finite.

Assumption 1 is not a modeling choice; it is what the exchangeability argument requires. If $(A,\kappa)$ were refined using the calibration labels of the points being predicted, the calibration and test scores would no longer be exchangeable and the distribution-free guarantee would fail. Sample splitting, in which the design is fit on one fold and frozen for another, is the standard way to satisfy Assumption 1, and it is still a frozen design.

Write $W = A(X,Y)$ for the score as a random variable and $\kappa = \kappa(X)$ for the stratum. Let $P_W(\cdot\,|\,x)$ be the true conditional law of $W$ given $X = x$, and let $G_\kappa = P_W(\cdot\,|\,\kappa(X) = \kappa)$ be the stratum-pooled score law, the population object that $\widehat{G}_\kappa$ estimates.

**Theorem 3** (Irreducible floor)**.** *Under Assumptions 1 and 2, the excess risk of the conformal predictor* $\widehat{F}_n$ *of Definition 1 satisfies*

$$L(\widehat{F}_n) - H(Y \mid X) \;=\; \mathrm{I}\big(W; X \mid \kappa\big) \;+\; \Delta_n, \qquad \Delta_n \to 0, \tag{3}$$

*where* $\mathrm{I}(W; X \mid \kappa) = \mathbb{E}_X \operatorname{KL}\big(P_W(\cdot \mid X) \,\|\, P_W(\cdot \mid \kappa(X))\big)$ *is the conditional mutual information between the score and the input given the stratum, and* $\Delta_n = \mathbb{E}_X \operatorname{KL}\big(G_{\kappa(X)} \,\|\, \widehat{G}_{\kappa(X)}\big)$ *is the pooled-density estimation term, which vanishes as the calibration size in each stratum grows.*

*Proof.* Fix the fold, so $(A, \kappa)$ is deterministic. By Assumption 2 the score $W = A(x, Y)$ has, given $X = x$, the density obtained by change of variables, $p(y \mid x) = p_W\big(A(x, y) \mid x\big)\, A'(x, y)$. The conformal density is $\widehat{f}_x(y) = \widehat{g}_{\kappa(x)}\big(A(x, y)\big)\, A'(x, y)$ with $\widehat{g}_\kappa$ the density of $\widehat{G}_\kappa$. The Jacobian $A'(x, y)$ is common to both, so it cancels in the log ratio:

$$\log \frac{p(y \mid x)}{\widehat{f}_x(y)} \;=\; \log \frac{p_W\big(A(x, y) \mid x\big)}{\widehat{g}_{\kappa(x)}\big(A(x, y)\big)}.$$

Taking expectations over $Y \mid X = x$ and substituting $w = A(x, y)$,

$$\operatorname{KL}\big(P(\cdot \mid x) \,\|\, \widehat{F}_x\big) \;=\; \operatorname{KL}\big(P_W(\cdot \mid x) \,\|\, \widehat{G}_{\kappa(x)}\big).$$

Average over $X$ and use (1). Split the divergence at the population pooled law $G_{\kappa(x)}$:

$$\operatorname{KL}\big(P_W(\cdot \mid x) \,\|\, \widehat{G}_{\kappa(x)}\big) = \operatorname{KL}\big(P_W(\cdot \mid x) \,\|\, G_{\kappa(x)}\big) + \mathbb{E}_{W \sim P_W(\cdot \mid x)}\left[\log \frac{g_{\kappa(x)}(W)}{\widehat{g}_{\kappa(x)}(W)}\right].$$

The second term averages over $X$ to $\Delta_n$. For the first, because $\kappa(x)$ is a function of $x$ we have $P_W(\cdot \mid X, \kappa) = P_W(\cdot \mid X)$ and $G_\kappa = P_W(\cdot \mid \kappa)$, so

$$\mathbb{E}_X \operatorname{KL}\big(P_W(\cdot \mid X) \,\|\, P_W(\cdot \mid \kappa(X))\big) \;=\; \mathrm{I}\big(W; X \mid \kappa\big)$$

by the definition of conditional mutual information. This gives (3). The term $\Delta_n$ is the expected excess of the log-loss of the estimated pooled density over the true pooled density within a stratum; under Assumption 1 the calibration labels are exchangeable within each stratum, so $\widehat{G}_\kappa \to G_\kappa$ and $\Delta_n \to 0$ as each stratum fills. □

**Corollary 4** (Positivity and suboptimality)**.** $\mathrm{I}(W; X \mid \kappa) > 0$ *unless* $W \perp X \mid \kappa$*, that is, unless the frozen pair* $(A, \kappa)$ *renders the score conditionally independent of the input within strata. When it does not, the excess risk is bounded away from zero for every* $n$*:* $\liminf_n \big[\, L(\widehat{F}_n) - H(Y \mid X) \,\big] \geq \mathrm{I}(W; X \mid \kappa) > 0$.

# 4 What calibration cannot do

The floor $\mathrm{I}(W; X \mid \kappa)$ is a functional of the frozen pair $(A, \kappa)$ alone. The calibration data enter Theorem 3 only through $\Delta_n$, which vanishes. The ranking, quantile, and $p$-value machinery of conformal prediction is precisely the computation of $\widehat{G}_\kappa$ and its quantiles, so it moves $\Delta_n$ and nothing else.

**Proposition 5** (The calibration step is a floor no-op)**.** *Under Assumptions 1 and 2, no choice of calibration estimator, quantile convention, randomization, or level reduces the excess risk below* $\mathrm{I}(W; X \mid \kappa)$*. Any reduction of the floor is a change to* $A$ *or* $\kappa$*.*

*Proof.* Immediate from (3): the first term does not depend on the calibration labels once $(A, \kappa)$ is fixed, and every conformal calibration operation is a function of the pooled scores $\{A(Z_i)\}$, hence acts only through $\widehat{G}_\kappa$ and $\Delta_n$. $\square$

Proposition 5 settles the schemes that appear to escape the characterization by acting after the pooled law is formed. Adaptive conformal inference (Gibbs and Candès, 2021) adjusts the level $\alpha_t$ online from past coverage; cross-conformal and CV+ average pooled laws across folds. Both are functions of the calibration scores alone, so both move only $\widehat{G}_\kappa$ and its quantiles, and by Proposition 5 neither reduces $\mathrm{I}(W; X \mid \kappa)$. Level adaptation and fold averaging are calibration operations; the floor is a design quantity.

The reason the floor is positive in practice is sharp enough to state as a remark.

*Remark* 6 (The use case is the floor). Call $(A, \kappa)$ *conditionally sufficient* if $W \perp X \mid \kappa$. By Corollary 4 the floor is zero exactly when the frozen design is conditionally sufficient, and a conditionally sufficient design is one whose pooled residual law already is the conditional law, that is, one for which the base modeling was finished. Conformal calibration is deployed when one is unwilling to assume the residual law is conditionally adequate. That unwillingness is the negation of conditional sufficiency, and by Corollary 4 it is the condition under which the floor is strictly positive. The method's stated reason for existing is its lower bound.

## 5 Full conformal

In full conformal prediction the score is refit for each candidate, so $A$ depends on the augmented multiset $\{Z_1, \dots, Z_n, (x, y)\}$ through a function that is symmetric in its arguments. The reshaping $A(x, \cdot)$ is then random, but it remains measurable with respect to the data and, crucially, the symmetry is what supplies exchangeability rather than a separate fold.

**Lemma 7** (Full-conformal floor)**.** *Let $A_n$ be a nonconformity score, symmetric in the calibration points and, for fixed $x$, a strictly increasing absolutely continuous bijection in $y$. Assume the following design stability: there is a limiting score $A_\infty(x, \cdot)$ and a limiting pooled law $G_{\kappa(x)}$ with $A_n(x, \cdot) \to A_\infty(x, \cdot)$ and $\widehat{G}_{\kappa(x)} \to G_{\kappa(x)}$ almost surely, and the score densities are uniformly integrable so that the divergences converge. Then the conclusion of Theorem 3 holds with $A_\infty$ in place of $A$: the limiting excess risk is* $\mathrm{I}(A_\infty(X, Y); X \mid \kappa)$.

*Proof sketch.* Condition on the calibration multiset. Symmetry makes the augmented scores exchangeable, so the rank of the test score is uniform and (2) holds with the realized $A_n$. The change of variables in Theorem 3 uses only that $A_n(x, \cdot)$ is a strictly increasing absolutely continuous bijection for fixed $x$, which holds pathwise; the Jacobian cancels as before. Passing to the limit with $A_n \to A_\infty$ and $\widehat{G}_\kappa \to G_\kappa$, dominated by Assumption 2, gives the stated floor. The only new ingredient beyond Theorem 3 is that the frozen design is replaced by a data-measurable but label-symmetric design, which preserves both the reshaping form and the cancellation. $\square$

Lemma 7 closes the characterization: whether the design is fold-frozen or refit-symmetric, the predictive is a reshaping of a pooled law and the floor is a conditional mutual information of the score.

## 6 The rate handicap of consistent schemes

Some schemes attack the floor directly by refining the stratification. The universally consistent predictive system of Vovk (2022) takes $\kappa = \kappa_n$ to be a dyadic partition with cells of width $h_n \to 0$, so that $\mathrm{I}(W; X \mid \kappa_n) \to 0$ and the excess risk vanishes. Consistency is genuine. Efficiency is not.

**Corollary 8** (Regressogram handicap)**.** *Let $\kappa_n$ be a data-independent refinement with $|\mathcal{K}_n| \to \infty$. Then Assumption 1 requires each cell to retain growing calibration mass, $n/|\mathcal{K}_n| \to \infty$, for $\Delta_n \to 0$, and forbids choosing the cells from the calibration labels. The resulting estimator is a fixed-design, label-blind regressogram, whose excess risk decays at the fixed-design nonparametric minimax rate. Where the conditional law has spatially inhomogeneous complexity, this rate is strictly dominated by the label-adaptive minimax rate attained by an estimator permitted to place resolution using the labels (Stone, 1982; Donoho and Johnstone, 1998).*

The handicap has the same cause as the floor. Assumption 1 says the labels may not choose where resolution is spent, so the partition cannot be refined where the truth is rough and coarsened where it is smooth. A fixed partition pays the worst-case local complexity everywhere. The consistent conformal scheme is a regressogram by construction, and a regressogram is minimax-suboptimal wherever adaptation would help.

## 7 Discussion

The two frozen ingredients $(A, \kappa)$ are a coarsening of the input. Everything a conformal scheme can represent about the conditional law lives in that coarsening; the calibration step estimates the pooled score law within it and nothing more. The excess risk is therefore governed by a single quantity, the conditional mutual information $\mathrm{I}(W; X \mid \kappa)$ that the coarsening leaves behind, and this quantity is fixed before any calibration is done. Three consequences follow, and together they exhaust the options.

If the coarsening carries no input dependence in the shape, one stratum and a location score, the floor is the plain residual information gap $\mathrm{I}(R; X)$ (Cotton, 2026), and no data reduce it. If the coarsening uses the input but is frozen, normalized, quantile, or Mondrian, the floor is the smaller but still positive $\mathrm{I}(W; X \mid \kappa)$, positive by Remark 6 exactly when the modeling was not finished. If the coarsening refines toward sufficiency, the labels may not steer the refinement, and Corollary 8 charges the regressogram rate. In each case the conformal calibration step contributes zero to the floor, by Proposition 5. What reduces the floor is a better $(A, \kappa)$, which is ordinary conditional modeling done before the conformal step, and which the change of variables shows the conformal step then merely reshapes and certifies.

This does not diminish the coverage certificate, which is exact at its position and distribution-free. It locates it. The certificate is a statement about the marginal rank of the test score; the floor is a statement about the conditional information the frozen score discards. The two live on different axes, and the first cannot move the second, which is why a valid conformal predictor and a sharp one are related only through the modeling that precedes the calibration.

## References


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